

The law of reflection

Euclid: *Catoptrica*

Def.3 → Prop.1 : similar triangles

Scholium 7 (ascribed to Archimedes) : symmetry

Pseudo-Ptolemy (Hero ?) : *De speculis*

Prop.4: the principle of minimum lines

Ptolemy: *Optica*

III-3ff.: (a) an empirical fact (cf. Archimedes)

(b) an experiment

(c) a theoretical reasoning

Al-Kindi: *De aspectibus*

Prop.16: an experiment

Prop.17: physico-mathematical consideration

Alhazen: *Kitāb al-manāẓir* (*De aspectibus* or *Perspectiva*)

IV-7: decomposition of motion into a perpendicular and a horizontal components and
their preservation (like Descartes's proof)
: an experiment

R. Grosseteste: *De luce*

De lineis, angulis et figuris

R. Bacon: *Perspectiva* (the Part V of the *Opus maius*)

De multiplicacione specierum

the natural philosophical proof

Witelo: *Perspectiva* (cf. Alhazen)

J. Pecham: *Perspectiva communis* (cf. Grosseteste & Bacon)

G. Valla: *De expetendis et fugiendis ...*

Free translation of Euclid's *Catoptrica*

symmetry (Archimedes) and similar triangles (Euclid's Prop.1)

F. Maurolico: *Photismi de lumine et umbra ...*

Supp.3: uniqueness of the path of reflection

Supp.4: principle of reciprocity

Theorems 26 & 27: symmetry (Valla's influence ?)